Presentation Script

Impact of Weather Information Availability on One-Hour-Ahead Bike-Sharing Demand Forecasting: A Cross-Model Study

Predictive Learning · Midterm Research Proposal · October 27, 2026

Speaker plan (14 slides, ≈ 12–14 minutes)

Speaker 1	Slides 1–4	Opening, motivation, problem, related work
Speaker 2	Slides 5–7	Weather evidence, research gap, aims and RQs
Speaker 3	Slides 8–10	Data, features, information settings, X1–X3
Speaker 4	Slides 11–14	X4–X5, evaluation, outcomes, limitations, closing

Italic orange lines are hand-off sentences that link one slide to the next. Written in a plain graduate-student speaking style; adjust the wording to your own habits.

SPEAKER 1 — Slides 1–4

Slide 1 · Title

Hello everyone, we are Group [X]. Our topic is about one-hour-ahead bike-sharing demand forecasting, and we focus on weather information. Basically, when we predict the next hour, we want to know which weather data we can really use at that time, and whether it improves the forecast. I will introduce the background and the research problem first. Then my groupmates will talk about the research gap, the experiment design, and the expected outcomes.

Slide 2 · Research Background and Motivation

First, the background. Many studies show that weather affects bike-sharing use. For example, Gebhart and Noland found that cold temperature, rain and high humidity reduce both the number and the length of trips. Eren and Uz also reviewed the literature, and weather and time factors appear in many studies. For data, we use the Seoul Bike Sharing Demand dataset, which has hourly rental counts with weather and calendar information. Sathishkumar and Cho already used it for weather-based prediction. However, a forecast can only use information that is already known when we make it. So we think the evaluation should follow time order: train on earlier months and test on later months. Also, our scope is only the system-wide hourly rental count. We do not have station data, so we do not make any claim about rebalancing.

Next I will explain the research problem more clearly.

Slide 3 · Research Problem and Forecasting Scenario

Our main question is which information is available at the forecast origin, and how complex the model should be. The target is the system-wide rented bike count in the target hour. The forecast origin is the end of the previous hour, so we can only use earlier observations. We do rolling one-step forecasts, and the model parameters are fixed during testing. From this we have three problems. P1 is information availability. Is the previous hour's observed weather useful, and how much does the actual weather of the target hour change the result? P2 is model complexity. How do regularization and flexibility affect generalization, when we train on winter and spring and test on later months? P3 is attribution and errors. What does each part add, including demand history, time, calendar and weather, and when does the forecast fail?

Before our approach, I will go through the related work.

Slide 4 · Related Work: Bike-Sharing Demand Forecasting

This table shows three main studies. Eren and Uz is a review of station-based bike-sharing research. They organize the factors of demand, such as weather, built environment, public transport, socio-demographics, temporal and safety factors. Sathishkumar and Cho build a rule-based model with weather data for Seoul, and they use the same dataset as ours. Xin, in 2026, does one-hour-ahead forecasting for Seoul and Washington, DC with Transformer-based and other models, and reports the point error and the 80% interval coverage. So from these works, we can see that weather is linked to demand, and Seoul hourly demand has been forecasted with learned models.

Then my groupmate will continue with the weather part and the gap.

SPEAKER 2 — Slides 5–7

Slide 5 · Related Work: Weather Information and Predictive Models

Thank you. On this slide, the left side is the evidence about weather. It is the same studies: Gebhart and Noland in Washington, DC, the review by Eren and Uz, and Sathishkumar and Cho in Seoul. The right side is the result of Xin on Seoul, one hour ahead. N-BEATS and Random Forest have an MAE of about 115 rentals per hour, while TFT and Transformer are around 307 and 318. For the coverage, Random Forest is 89% and N-BEATS is 75%, where the target is 80%. So the model with the lowest point error does not always have the best coverage. These numbers are from Xin's paper, not our results, so they are not directly comparable with ours. One thing that is still unclear is which weather values are really available at the forecast origin for a one-hour-ahead forecast. The papers do not describe it field by field.

This is the part we want to address.

Slide 6 · Research Gap and Proposed Contribution

So we summarize three gaps. The first is information timing. Xin uses target-time covariates, but whether each field is available at the forecast origin is not fully specified. We are not saying there is leakage in that work. What we do is define three information settings, E1, E2 and E3, under the same protocol. The second is complexity control. Previous work compares models from different families, but does not focus on controlled complexity curves. So we make the curves for Ridge alpha, KNN k and tree depth, and compare training and validation error. The third is information value. It is not clear which input group matters most, or where the errors are concentrated. So we do group ablation and temporal error analysis. We use the same split dates as Xin, but we use five weather variables, three demand lags, a 168-hour warm-up, and our own definition of eligible samples. Also, we do not claim to be the first weather study, we do not claim leakage in prior work, and we do not claim to be better than probabilistic methods.

Based on this gap, here are our aim and research questions.

Slide 7 · Research Aim and Research Questions

Our aim is to separate the value of available weather information from retrospective information, and to study model complexity under out-of-time evaluation. We have three research questions, and each one has a hypothesis that can be tested. RQ1 asks whether the previous hour's observed weather improves the next-hour forecast, and how the target hour's actual weather is different. We expect E2 may be better than E1, and E3 may help in precipitation hours, but it is not guaranteed to be better than E2. RQ2 asks how complexity and regularization affect out-of-time generalization. We expect moderate regularization may help, and a complex model does not necessarily win. RQ3 asks what demand, time, calendar and weather contribute, and where the models fail. We expect demand history to add the most, and the peak hours to have larger absolute errors. The last column shows which analyses, X1 to X5, are used for each question.

My groupmate will now introduce the data and the setup.

SPEAKER 3 — Slides 8–10

Slide 8 · Dataset, Preprocessing and Chronological Split

Thank you. For the dataset, we use the Seoul Bike Sharing Demand data. It has 8,760 hourly observations and 14 columns, and there are no gaps or missing values. After preprocessing, we have 8,297 eligible samples. The bar shows the chronological split. December 1 to 7, 2017 is the 168-hour warm-up. Training is from December 8 to the end of May, 4,152 samples. Validation is June, 720 samples. Test is from July 1 to November 30, 2018, 3,425 samples. For preprocessing there are four steps. First, we combine date and hour into a timestamp. Second, we build the lag features, lag_1, lag_24 and lag_168, on the full hourly axis before selecting rows. Third, we only score the hours with Functioning Day equal to Yes and with all lags available. Fourth, the scaler and encoders are fitted on the training rows only. For leakage control, we use a chronological split, lags from the past only, and we never use the target hour's demand as an input.

Next is the feature design.

Slide 9 · Feature Engineering and Information Settings (E1–E3)

This figure shows the features and the three information settings. The dashed line separates the past and the target hour. We have four feature groups. H is historical demand: lag_1, lag_24 and lag_168. T is time, which is the hour and weekday, encoded with sine and cosine. C is calendar, which is season and holiday. W is weather, with five variables: temperature, humidity, wind speed, rainfall and snowfall. For the settings, E1 uses H, T and C, so there are 13 features and no explicit weather. E2 adds the observed weather of the previous hour, which is available at the forecast origin, so it has 18 features. E3 replaces it with the weather of the target hour. This is a retrospective oracle, so it cannot be deployed, and it does not need to beat E2. In all settings, the target-hour demand is not used.

Then I will explain how we use these settings in the experiments.

Slide 10 · Experimental Design: X1–X3

We have five analyses, X1 to X5, and they are crossed with E1 to E3, so they are not eight separate experiments. This table shows the models and the settings for each analysis. X1 is the model comparison. We use Ridge, KNN and Random Forest under E1, E2 and E3, together with three naive baselines, to see whether the model ranking changes with the information setting. X2 is Ridge regularization. We fix E2, use ten log-spaced alpha values, and compare the training and validation RMSE. The objective function is shown at the bottom. X3 is complexity control. E2 is still fixed, and we change k of KNN from 1 to 97, and the max depth of the decision tree from 2 to 16 and None. We want to check whether more flexibility only reduces the training error.

Now X4 and X5 will be introduced by my groupmate.

SPEAKER 4 — Slides 11–14

Slide 11 · Experimental Design: X4–X5

Thank you. X4 is about weather information and feature ablation. In 4a, we compare E1, E2 and E3 with Ridge and Random Forest, using the same samples. We define Gain_history as the MAE of E1 minus the MAE of E2, which is the benefit of the available weather, and Gap_oracle as the MAE of E2 minus the MAE of E3, which is what the retrospective weather could add. In 4b, we remove one group from the full E2 model at a time, which is H, T, C or W, keep the hyperparameters fixed, and see how much the MAE changes. Removing W is the same as E1. X5 is the temporal error analysis. We use the best E2 model selected by validation MAE, and look at the test period in slices that are decided in advance. These are by hour, including night and peak hours, by month, weekday and weekend with the holiday flag, and rain or snow versus none. We also plot the actual and predicted curve for a fixed week. Note that ablation is not a causal intervention.

Next is the evaluation protocol.

Slide 12 · Evaluation Protocol, Resources and Project Timeline

For the metrics, we use MAE and RMSE in rentals per hour, and the predictions are clipped at 0. We do not use MAPE because there are zero counts. Validation MAE is used to select the hyperparameters, and the test set is evaluated only once with a list decided in advance. The scalers are fitted on the training rows only, and we do not use shuffled K-fold. For the timeline, we upload to Moodle on October 26 and present on October 27. In week 1, we build the shared pipeline, the baselines and Ridge. In week 2, we run X1 to X3 on train and validation and freeze the settings. In week 3, we set up X4 and do the LSTM extension. In week 4, we freeze the list and run the test and X5 once. In week 5, we check reproducibility and finish the final report. We use Python, pandas, scikit-learn and matplotlib on laptop CPUs, so there is no additional budget. For the work distribution, one person does the data and pipeline, one does Ridge and KNN, one does Random Forest, and one does X5, the integration and the LSTM extension. The data is public and aggregated, so there is no human subject or personal data, and we do not claim any ethics review.

Finally, the outcomes and limitations.

Slide 13 · Proposed Outcomes and Research Limitations

For the outcomes, we plan to provide a table of MAE and RMSE by model and information setting with the naive baselines, the Gain and Gap between E1, E2 and E3, the curves of Ridge alpha, KNN k and tree depth, a group-ablation chart, the error analysis by hour, month, weekday and precipitation, and a reproducible pipeline that is checked for leakage. For the limitations, we only use one city and one year, so there is no multi-year or cross-city claim. Season and time are confounded in the split. Aggregate rentals are not station demand or unmet demand. E2 assumes the previous hour is already published, and E3 is oracle information, not a real weather forecast. We also only do point forecasts, without interval calibration. The figures in the final report will come from executed experiments, and we do not claim any result now. To conclude, our contribution is a reproducible protocol that separates available and retrospective weather information, with complexity and error analysis. That is the end of our presentation. Thank you, and we are happy to answer your questions.

Slide 14 · References (shown during Q&A)

This slide is for reference during the Q&A. It lists the papers and datasets we mentioned: Eren and Uz, Gebhart and Noland, scikit-learn by Pedregosa and others, Sathishkumar and Cho, Xin, and the UCI and Kaggle versions of the dataset. Thank you.